

Arthur Jiun Wei Hwang

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Mechanical Engineering

Sep. 2024 - May 2029 (Expected)

- Professional Experience Year (PEY) Co-op Program
- 3.97 Cumulative GPA, Dean's Honour List 2024 & 2025
- Minor in Robotics & Mechatronics and Engineering Business, Certificate in Electric Vehicle Design

EXPERIENCE

RocketPy Open-Source

Remote

Library Development Member

Dec. 2024 - Present

- Maintaining a Python library using Git version control, proper documentation, testing, and code review practices
- Improved parachute aerodynamic model accuracy by modeling it as a hemispheroid parameterized by radius, height, and porosity
- Developed serialization functions to save and quickly retrieve Flight simulation data via custom RPY file format

Laboratory for Extreme Mechanics & Additive Manufacturing

Toronto, ON

Summer Research Student

May 2025 - Aug. 2025

- Developed G.L.O.W., an open-source GCODE generator desktop application for Direct Energy Deposition Additive Manufacturing, supporting 3 print geometries, 10 print parameters, and 11 preprogrammed machine-specific codes
- Ported G.L.O.W. from Python to C++ using Qt Creator, reducing binary size from 1.6 GB to a 270 MB lightweight executable and improving runtime performance by 40%, enabling practical deployment across lab workstations
- Presented G.L.O.W. and GCODE generation methodology as guest speaker in graduate-level MSE1068 Additive Manufacturing course at University of Toronto

EXTRACURRICULAR

UAS Division, University of Toronto Aerospace Team

Toronto, ON

SAE Structures Lead

Jun. 2026 - Present

- Led a multidisciplinary team of 8 in the use of computation and simulation tools (XFLR5, FLOW5, SolidWorks, MATLAB) to design a lifting-body fuselage for an autonomous fixed-wing/VTOL tricopter under 2 months
- Integrated the avionics bay, wing attachment, and tail mount into a lightweight load-bearing architecture using strategic lightening-hole layout to reduce mass by 58.75% and meet the 150g mass constraint
- Optimized landing gear placement within size constraint to provide 90-10% weight distribution, easing take-off pitching motion and reducing elevator control input

SAE Aerodynamics Member

Sep. 2024 - Jun. 2026

- Selected lifting-body fuselage airfoil from 14 candidates via XFLR5 analysis, targeting stable CL and CD profiles at operating Reynolds numbers and sufficient internal volume to house avionics
- Designed modular assembly jigs that eliminated significant dimensional deviation in manufactured wings and resolved recurring fuselage alignment and wing warping issues from previous iterations

Rocketry Division, University of Toronto Aerospace Team

Toronto, ON

Aerodynamics Co-lead

Jan. 2026 - Present

- Directed a 5 member team in aerosurface design and flight analysis across three concurrent rocket programs, including the team's first double-stage solid rocket, first liquid rocket, and second hybrid rocket
- Designed and optimized nose cones and fins geometries in OpenRocket to achieve off the rail stability margins of 1.5 to 2 calibers for subsonic and 3 to 6 calibers for supersonic rockets across varying wind conditions (0 to 30 km/h)
- Modified and extended RocketPy codebase by implementing custom classes and overriding core methods to support multi-stage Monte Carlo dispersion simulations, enabling risk analysis for the team's first double-stage rocket

Aerodynamics Member

Sep. 2024 - Jan. 2026

- Automated Monte Carlo dispersion analyses for 4 rocket projects in RocketPy by modeling manufacturing tolerances and wind variability with normal distributions to predict probable landing zones for nominal, drogue-failure, and ballistic descent scenarios and verify safety compliance via KML export to Google Earth
- Developed a Jupyter Notebook to automatically build RocketPy simulations from OpenRocket files and scattered data sources (Excel, RSE files), reducing cross-validation tasks from 5+ hours to 4 seconds
- Implemented a Genetic Algorithm Multidisciplinary Optimization framework integrated with RocketPy to optimize aerosurface geometry, replacing manual parameter sweeps